

Quiz 2

ECON 308 – International Economic Relations

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Multiple-Choice Questions (1 point each)

1. In the specific-factors model, which factors gain unambiguously when the relative price of manufactures increases?
 - A) Landowners
 - B) Capital owners
 - C) Workers
 - D) None of the above
2. If P_M/P_F rises, what happens to the allocation of labor?
 - A) More labor moves into food
 - B) Labor allocation is unaffected
 - C) More labor moves into manufactures
 - D) Labor leaves both sectors
3. Why is the effect of a price change on labor's real wage ambiguous?
 - A) Because nominal wages do not change
 - B) Because wages always fall when prices rise
 - C) Because real wage rises in terms of one good and falls in terms of the other
 - D) Because labor is specific to one sector
4. Which of the following best summarizes the distributional effects of trade in the specific-factors model?
 - A) All factors gain from trade
 - B) The mobile factor gains unambiguously
 - C) Specific factors in the export sector gain, specific factors in the import sector lose, labor is ambiguous
 - D) Only capital gains from trade

5. Suppose Home exports manufactures under free trade. Which of the following must be true?

- A) Home has more land than Foreign
- B) Home's opportunity cost of manufactures is lower than Foreign's
- C) Home's wage is higher than Foreign's in all goods
- D) Home has higher productivity in both goods

Problem (15 points)

Home produces Manufactures (M) and Food (F). Labor is mobile between sectors, capital is specific to M , and land is specific to F .

Production functions:

$$Q_M = K^{1/3}L_M^{2/3}, \quad Q_F = T^{1/2}L_F^{1/2}, \quad L_M + L_F = L.$$

Endowments: $K = 64$, $T = 100$, $L = 90$.

Prices: initially $P_M = 2$, $P_F = 1$. Later, P_M rises to 3 while $P_F = 1$.

1. Derive the marginal products of labor in each sector.
2. Find the initial equilibrium allocation of labor (L_M, L_F) and the wage w .
3. Compute outputs Q_M, Q_F , the return to capital (r_K), and the return to land (r_T).
4. After the rise in P_M , determine qualitatively (without solving fully) how L_M, w, r_K , and r_T change. Identify the winners and losers.
5. Explain why labor's welfare effect is ambiguous, using the real wage in terms of each good.